

Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.50932	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	75.74116	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1334.9248	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	0.0	13.06	n/a	135.71366	1	0	0	0.964856	13.74
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	0.0	3.0	n/a	273.08588	2	0	0	0.945758	13.085
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.273435	0.273435	139.52028	0	0	0	0.0	0.0

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.50932.
- Accepted epoch count: 0.
- Fallback epoch count: 0.

- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 75.74116.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1334.9248.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:

- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 0.0.
- Train penalized gap: 8.958333.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 13.06.
- Held-out runtime ms: 135.71366.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 1.616667.
- Worst held-out instances:
- X-n275-k28: feasible=False, penalized gap=15.7, errors=["NameError: name 'isinstance' is not defined", 'missing 274 customers']
- X-n247-k50: feasible=False, penalized gap=14.3, errors=["NameError: name 'isinstance' is not defined", 'missing 246 customers']
- X-n223-k34: feasible=False, penalized gap=13.1, errors=["NameError: name 'isinstance' is not defined", 'missing 222 customers']
- X-n190-k8: feasible=False, penalized gap=11.45, errors=["NameError: name 'isinstance' is not defined", 'missing 189 customers']
- X-n176-k26: feasible=False, penalized gap=10.75, errors=["NameError: name 'isinstance' is not defined", 'missing 175 customers']

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 273.08588.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.

- Worst held-out instances:
- X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 139.52028.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary (Conservative)

Aspect	Summary
Direct Synthesis	"direct_generate_plus_one_repair" showed 0% feasibility and very high penalized gap (~13.06), with high complexity (13.74) and low robustness.
Budget-Matched Independent Search	"budget_matched_no_replay" also had 0% feasibility and moderate penalized gap (3.0) at runtime 273 ms, complexity 13.1, no robustness dispersion (0).
Replay-Aware Iterative Search	"replay_solver_evolution" achieved 100% feasibility but had heldout gap of 0.273, similar to lowest baseline gaps but no superiority in objective gap. Runtime 140 ms, zero complexity/novelty, low robustness dispersion.

Aspect	Summary
Fixed-Baseline Context	Baselines tested: nearest_neighbor, clarke_wright_savings, and regret_insertion_local_search. Clarke-Wright savings had best heldout gap (0.0738), 100% feasibility, moderate runtime (76 ms), and low robustness dispersion. Other baselines had higher gaps or runtimes.
Feasibility	Only fixed baselines and replay-based search guaranteed 100% feasibility on heldout data; learned or budget-matched no-replay failed feasibility completely.
Objective Gap	Clarke-Wright baseline had the lowest feasible gap (0.0738), outperforming learned and replay variants. Replay (0.2734) matched nearest_neighbor baseline gap, worse than Clarke-Wright.
Runtime	Baselines ranged from ~43 ms (nearest_neighbor) to 1335 ms (regret_insertion). Replay at 140 ms was faster than regret local search but slower than Clarke-Wright. Learned methods were slower or infeasible.
Robustness Across Held-Out Families	Clarke-Wright and nearest_neighbor showed low robustness dispersion (~0.003-0.008). Replay solver had similar robustness. Learned methods exhibited high complexity and novelty but low robustness.
Learned Conditions vs. Fixed Baselines	None of the learned or replay-aware conditions clearly outperformed the best fixed baseline (Clarke-Wright savings) on objective gap or feasibility. Learned methods failed feasibility on held-out instances.
Replay vs. Budget-Matched No-Replay Control	Replay (100% feasibility, gap 0.273) outperformed budget-matched no-replay (0% feasibility, gap undefined) in feasibility and objective reliability, but neither surpassed baselines.

****Overall conclusion:****

The fixed Clarke-Wright savings baseline remains the best performing method in the phase-9 CVRP closeout suite on feasibility, objective gap, runtime, and robustness. Replay-aware iterative search improved feasibility relative to budget-matched no-replay learned search but did not beat the best fixed baseline. Direct synthesis and budget-matched independent searches failed to produce feasible solutions reliably.